Rigid prices

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Last time

A simple model of rigid prices

- Rational expectations
- The model produces a positive relationship between output and prices
- But policy makers cannot exploit it (unless they surprise everyone)

Lucas critique

- Don't take an empirical relationship for granted! Agents might reoptimize when we try to exploit it
- Expectations affect the equilibrium
- "Physicists do not teach atoms how to behave" - Luigi Zingales

Today

Identification of Phillips Curve with micro data

Price-setting Empirics

A New Keynesian model – light

- **Dynamic** model with monopolistic competition
- Main message remains intact: only unexpected shocks have effects

Price rigidities

- Fischer contracts in detail
- Taylor contracts intuition
- Menu costs intuition

Optimal policy

- What should central banks do against demand shocks?

Phillips Curve: Firm-level identification (Hoeck-Renkin, 2025)

Idea: Aggregate Phillips curve reflects

- *labor market pressure* (inelastic labor supply → wages ↑; last lecture)
- *goods market pressure* (inelastic capital → short-run capacity)
  → instrument firm demand with export shocks to recover the supply slope (identifies a part of the Phillips curve)

Shock (shift–share): $\Delta \tilde{z}_{i,t} = \sum_k \sum_j \omega_{i,k,j,t-1} \, \Delta \text{Imports}_{k,j,t}$

Lagged firm export shares $\omega_{i,k,j,t-1}$ over product j and destination k

Reduced form (LP): $\Delta p_{i,j,t+h} = \beta_h \, \Delta \tilde{z}_{i,t} + \text{sector} \times \text{time FE} + X_{i,t} + u_{i,t}$

Absorb home supply shocks and inflation expectations with sector×time FE

IV–LP for supply slope:

$\Delta p_{i,j,t+h} = \delta_h \, \Delta y_{i,t} + \text{FE} + X_{i,t} + e_{i,t}, \quad \Delta y_{i,t}$ IV: $\Delta \tilde{z}_{i,t}$

Key assumption: export shares $\gamma_{i,k} \perp$ idiosyncratic supply shocks v_i

Hoeck-Renkin, 2025

Local projection results

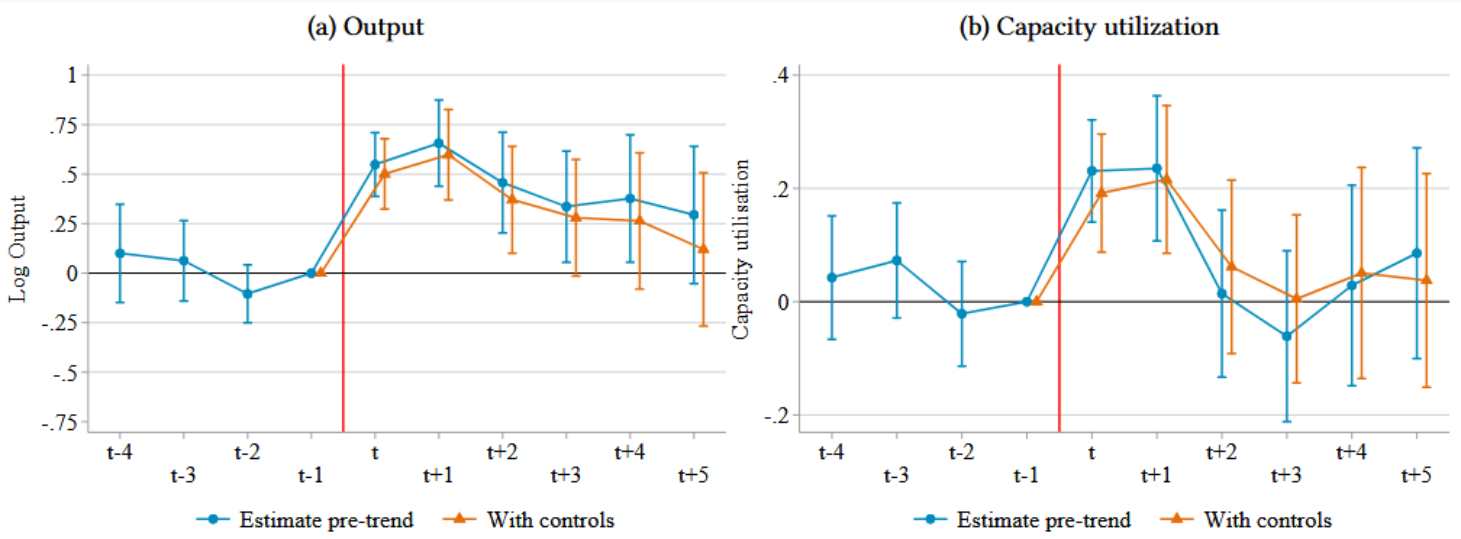


Figure 2: Effects of demand shocks on output and capacity utilization

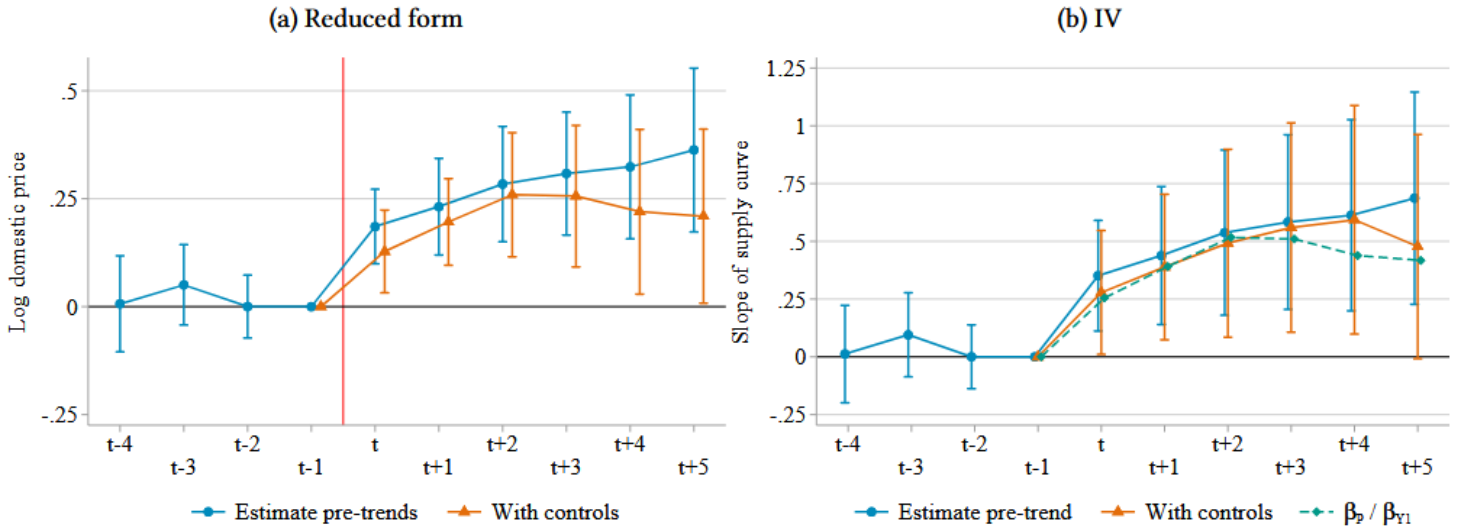


Figure 3: Effects on prices

Phillips Curve: Regional identification (Hazell et al., 2022)

Idea: Use U.S. states to learn about the Phillips curve

- All states share *one* central bank (the Fed)
  $\Rightarrow$ same monetary policy and long-run inflation regime
- But states differ a lot in local demand and unemployment
- Compare inflation and unemployment *across states*

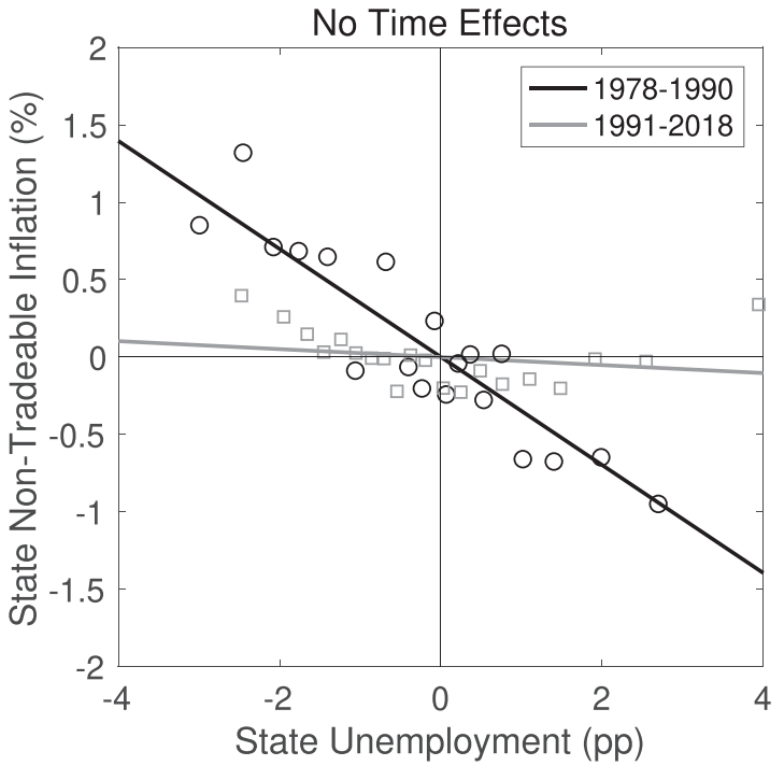
Data:

- New CPI data by U.S. state for *non-tradable* goods and services (e.g. haircuts, rent, restaurants)
- These prices are set locally, so they can respond to local slack

Empirical specification at state i level: $\pi_{it} = \alpha_i + \gamma_t - \psi\, u_{it} + \varepsilon_{it}$

- ψ: how much local inflation moves when local unemployment changes
- **IV:** Use national demand shocks to *tradable* industries as an instrument for state unemployment.
  $\rightarrow$ National booms in tradables (oil, cars, ...), weighted by a state's historical industry mix, raise local wages and thus costs in nontradables

Regional Phillips Curve



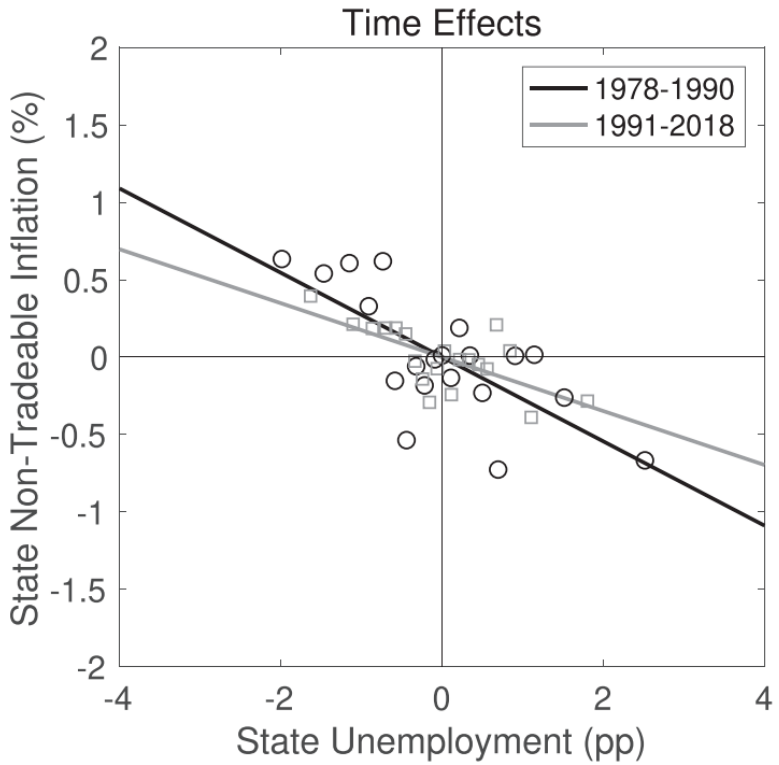


FIGURE V

Scatterplots: Nontradeable Inflation and Unemployment

Hazell et al., 2022

Summary: Identification of the Phillips Curve

- Identification of the Phillips curve is hard in aggregate time series
- A "cloud" of points in the aggregate inflation–unemployment scatter plot does *not* mean that there is no underlying Phillips curve
  → Clean identification of the slope requires demand shocks
- Micro and regional data can help identify parts of the Phillips curve:
  1. Firm-level demand shocks
  2. Regional heterogeneity across states
  3. ...

     → More work to do!

Empirics about price changes

Important research



Jon Steinsson & Emi Nakamura

- Five Facts about Prices: A Reevaluation of Menu Cost Models (2008)
- Price Rigidity: Microeconomic Evidence and Macroeconomic Implications (2013)

→ Need to know how we should model price rigidity

Data about price changes

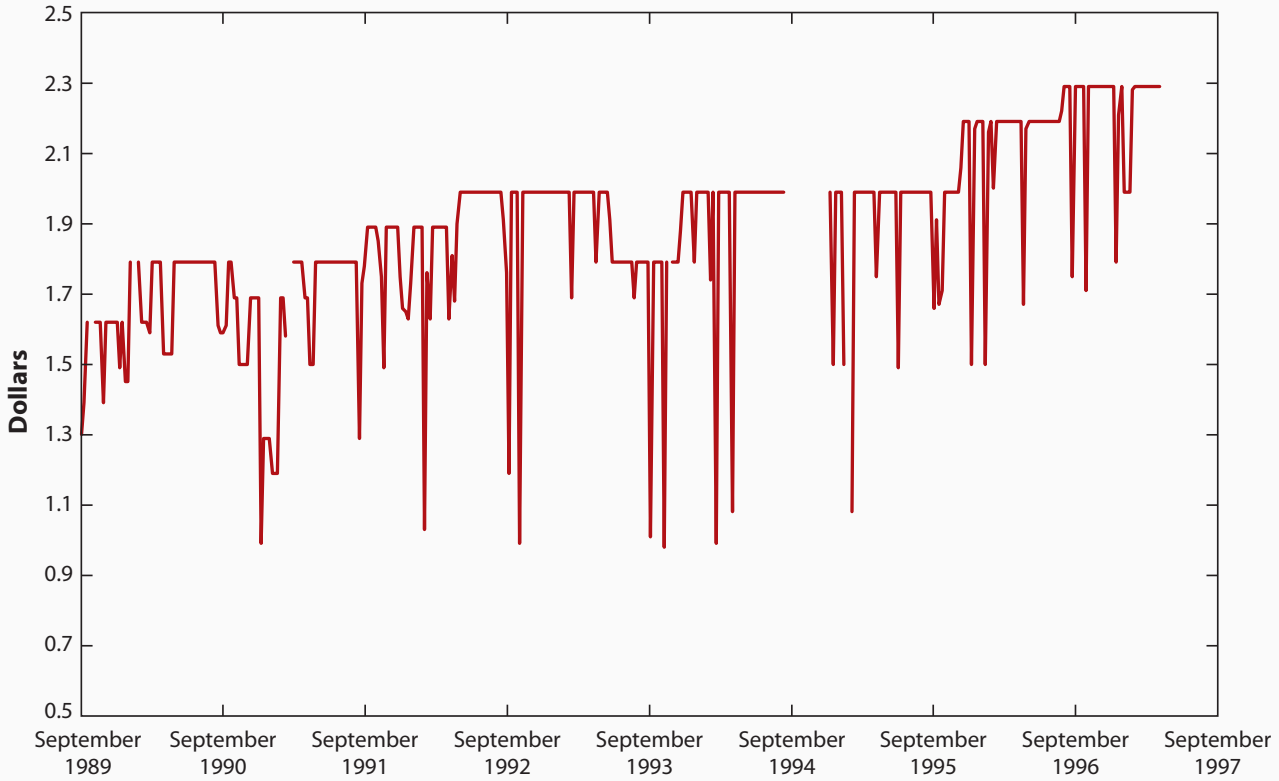


Figure 2

Price series of Nabisco Premium Saltines (16 oz) at a Dominick's Finer Foods store in Chicago.

- Is a sale a price change?

Data about price changes

		Regular prices			
		Median		Mean	
Major group	Weight	Freq.	Impl. dur.	freq.	Frac. up
Processed food	8.2	10.5	9.0	10.6	65.4
Unprocessed food	5.9	25.0	3.5	25.4	61.2
Household furnishing	5.0	6.0	16.1	6.5	62.9
Apparel	6.5	3.6	27.3	3.6	57.1
Transportation goods	8.3	31.3	2.7	21.3	45.9
Recreation goods	3.6	6.0	16.3	6.1	62.0
Other goods	5.4	15.0	6.1	13.9	73.7
Utilities	5.3	38.1	2.1	49.4	53.1
Vehicle fuel	5.1	87.6	0.5	87.4	53.5
Travel	5.5	41.7	1.9	43.7	52.8
Services (excl. travel)	38.5	6.1	15.8	8.8	79.0
All sectors	100.0	8.7	11.0	21.1	64.8

- Frequency of price changes is very heterogeneous
- Prices go up more than they go down

Data about price changes

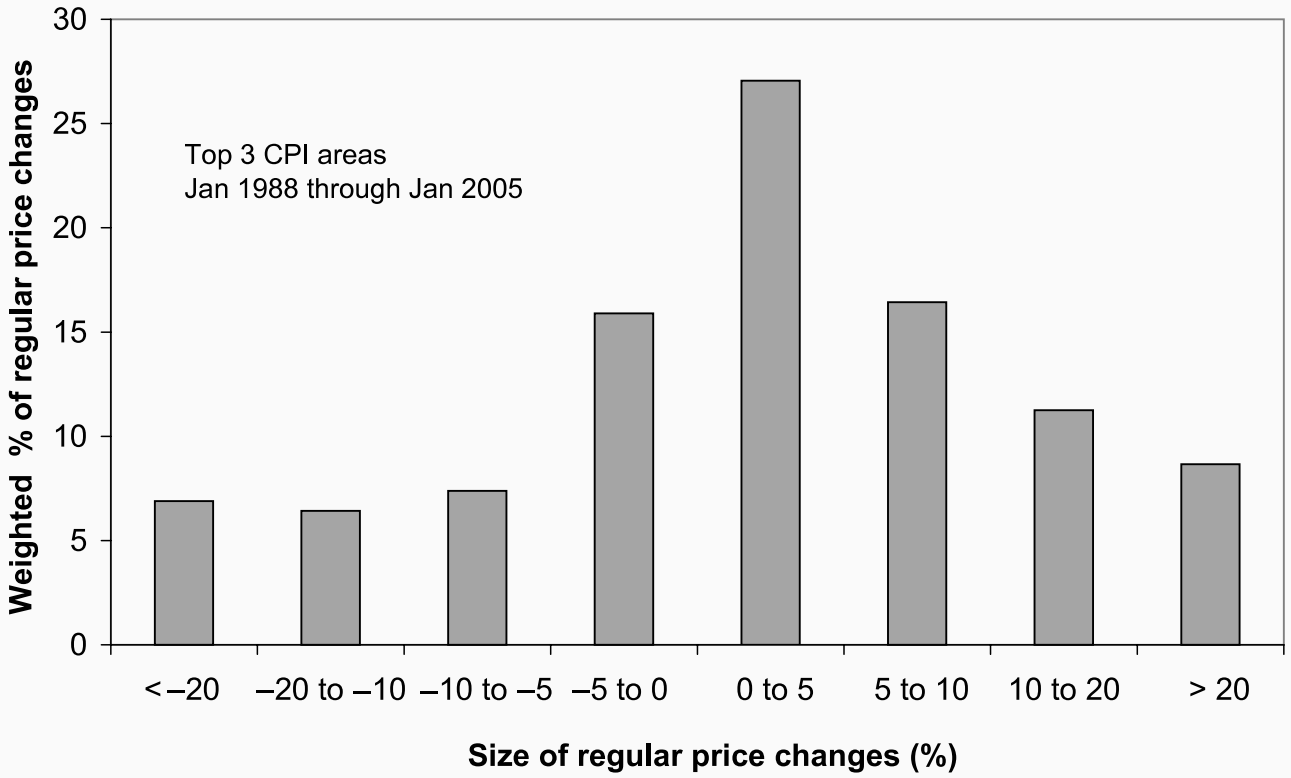


FIGURE II
Weighted Distribution of Regular Price Changes

- Average price change is positive and big $> 5\%$
- Many price changes are very small (Klenow & Krystov, 2008)

Data about price changes



FIGURE VIII
Size of Regular Price Changes by Age vs. Decile Fixed Effects

- Price change doesn't seem to depend on when price was last set
- Klenow & Krystov (2008)

Data about price changes

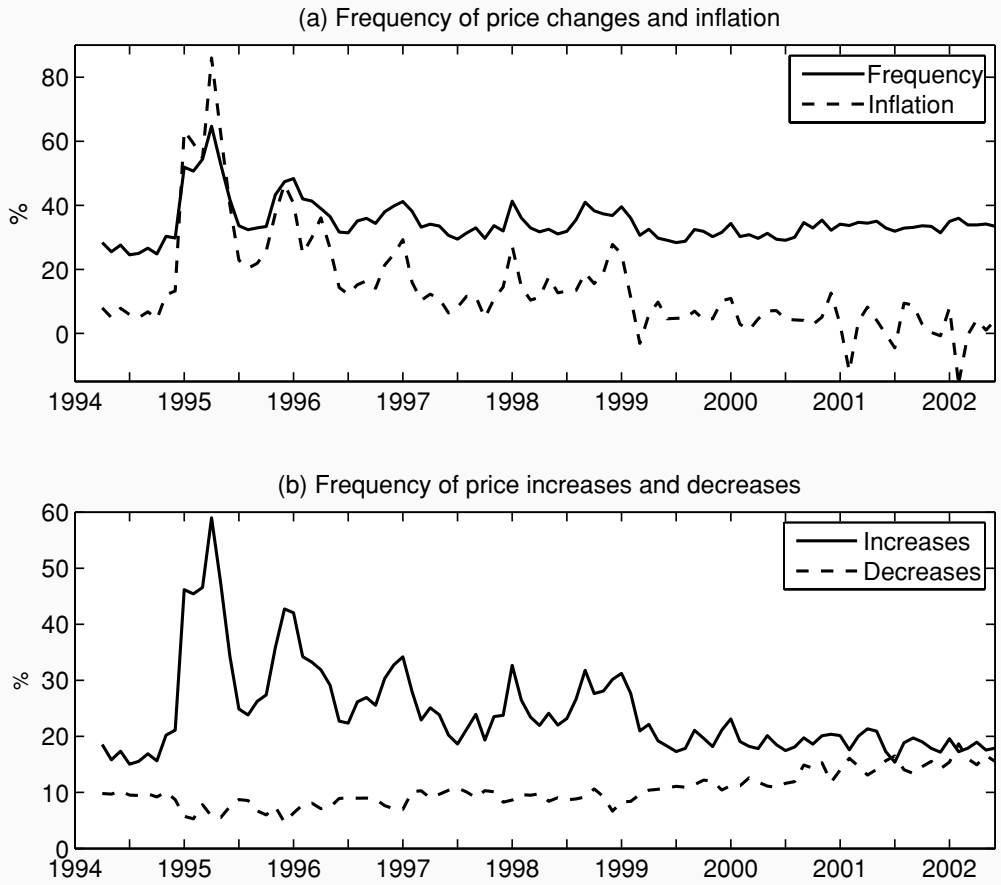


FIGURE III
Monthly Frequency of Price Changes (Nonregulated Goods)
All statistics in the figure, including inflation, are computed using the sample of nonregulated goods.

- In Mexico, high inflation means more adjustment (Gagnon, 2009)
- Frequency is flat for low inflation period → state dependence

Empirical summary

Price changes

- How often prices change depends on whether we count sales
- Most prices (without sales) change at least once a year
- In low inflation periods, most price changes are small
- Size of price change doesn't depend on how old the price is
- Higher inflation leads to higher and more frequent price changes

NK Model – light

The New Keynesian model – light

Today's model is almost the New Keynesian model

- As last time: cut out the middle man

→ No workers and firms, every household produces one good and consumes all goods

→ No wages, just disutility of work/utility of leisure

- Limited dynamics

Other crucial features

- Monopolistic competition and pricing power

→ **Keynesian**

Setup – same as last lecture

Representative household

$$U_i = \mathbf{C}_i - \frac{1}{\phi} L_i^{\phi} \quad \text{with } 1 < \phi$$

- $\mathbf{C}$ is a consumption basket, as in previous lectures, composed of C_i
- Households use their labor L to produce output according to $Y_i = L_i$
- P is the aggregate price level, P_i is the price of the household's variety i

Budget constraint

$$P\mathbf{C}_i = P_i Y_i \implies \mathbf{C}_i = \frac{P_i}{P} Y_i$$

Demand function

$$Y_i = \left(\frac{P_i}{P}\right)^{-\theta} Y \iff \frac{P_i}{P} = \left(\frac{Y_i}{Y}\right)^{-1/\theta} \quad \text{with } \theta > 1$$

Rearrange

Optimal output

$$\left(1-\frac{1}{\theta}\right)\left(\frac{Y_i}{Y}\right)^{-\frac{1}{\theta}} = Y_i^{\phi-1}$$

$$\underbrace{\left(\frac{\theta-1}{\theta}\right)}_{\text{Inverse markup}}\left(\frac{P_i}{P}\right) = Y_i^{\phi-1} \leftarrow \text{ p/P} = \text{markup} * \text{“marginal cost”}$$

Take logs

$$p_i - p = (\phi-1)y_i - \log\left(\frac{\theta-1}{\theta}\right)$$

$$p_i - p = (\phi-1)y_i + \mathcal{M}$$

Price setting

Optimal price setting

$$p_i^* - p = (\phi - 1)y_i + \mathcal{M}$$

- Optimal price depends on elasticity of labor supply $\frac{1}{\phi}$ and markup $\mathcal{M}$

Aggregate up (all firms are symmetric $\implies$ make the same choices)

$$p^* - p = (\phi - 1)\underbrace{(m - p)}_{y} + \mathcal{M}$$

$$p^* = (\phi - 1)m + (2 - \phi)p + \mathcal{M}$$

- Note that $p = m$ is not the outcome if $p^* = p$, due to monopolistic competition. Prices are higher in this model
- As ϕ rises (labor becomes less elastic) m's effect on p^* increases

Introducing expectations

Demand (money supply) is not deterministic

- Households must form expectations
- General expression! **How** m and p vary is irrelevant

$$p^*_{t-1} = (\phi - 1)\mathbb{E}_{t-1}[m_t] + (2 - \phi)\mathbb{E}_{t-1}[p_t] + \mathcal{M}$$

- Households form expectations about price level

$$\mathbb{E}_{t-1}[p_t] = \mathbb{E}_{t-1}\left\{(\phi - 1)\mathbb{E}_{t-1}[m_t] + (2 - \phi)\mathbb{E}_{t-1}[p_t] + \mathcal{M}\right\}$$

$$\mathbb{E}_{t-1}[p_t] = \mathbb{E}_{t-1}[m_t] + \frac{1}{\phi - 1}\mathcal{M}$$

Equilibrium

Prices and output

$$p = (\phi - 1)\mathbb{E}[m|I] + (2 - \phi)\left(\mathbb{E}[m|I] + \frac{1}{\phi - 1}\mathcal{M}\right) + \mathcal{M}$$

$$= \mathbb{E}[m|I] + \frac{1}{\phi - 1}\mathcal{M}$$

$$y = m - \mathbb{E}[m|I] - \frac{1}{\phi - 1}\mathcal{M}$$

- Only unanticipated movements in aggregate demand ($\mathbb{E}[m] \neq m$) have real effects
- Monopolistic competition leads to lower output and welfare loss
- Sometimes, we suppress the markup, but with Mon.Comp. it's there

Rigid/sticky prices

Various ways to model frictions in price-setting

Time-dependent pricing: timing of individual price changes is exogenous

- Fischer contracts: Set price **schedule** in advance
  → In detail today
- Taylor pricing: Fix prices for a certain time
- Calvo fairy: Fixed probability of adjusting → Next lecture

State-dependent pricing: firms choose when to change prices

- Menu costs: pay fixed price to change a price

Fischer contracts (Time-dependent, DR 7.2)

Environment

- Firms set price schedules in advance and stick to them
- Only some fraction of firms renews their schedules each period
- Everyone has to fulfil demand $\Longrightarrow$ work more if prices too low

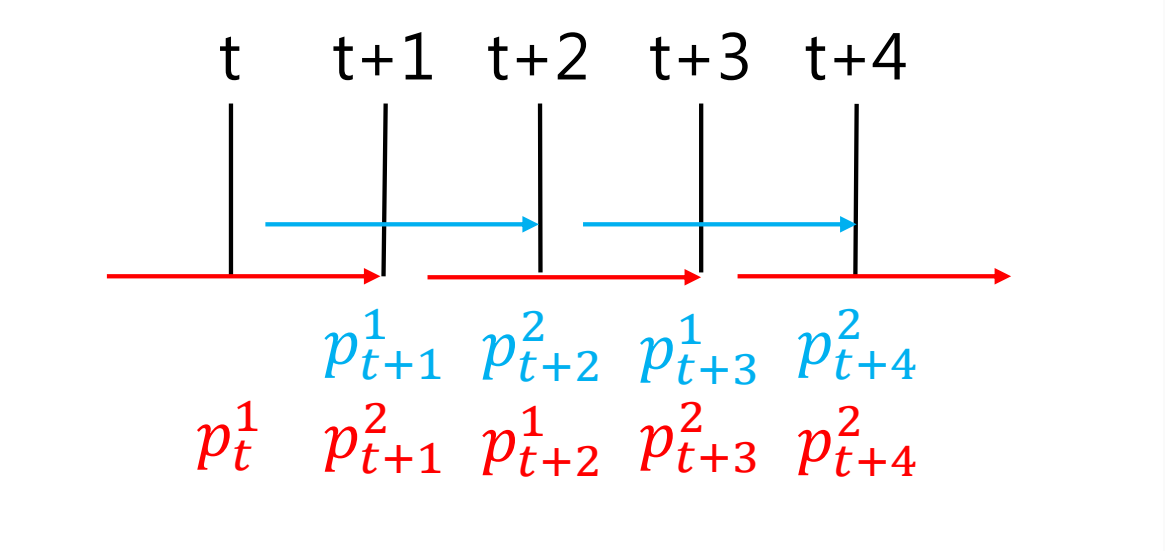
Operationalization

- Each price-setter sets prices for two periods – potentially different
- Assume that half of producers set prices in even periods, rest in odd
- Rational expectations: everyone knows the environment

Timing

- Those who reset do so right before the end of period $t-1$, setting prices for t and $t+1$
- They **do not** know the shocks in t

Timing and notation



- Price schedules are set with the information set of the previous period
- Prices can change over time
- Subscript: period for which prices were set
- Superscript: how many periods ago prices were set

Price level with Fischer contracts

Price level

$$p_t = \frac{1}{2}\left(p_t^1 + p_t^2\right)$$

$$p_t^* = (\phi - 1)m + (2 - \phi)p + \mathcal{M}$$

Optimal price setting in expectation

$$p_t^1 = \mathbb{E}_{t-1}[p_t^*] = (\phi - 1)\mathbb{E}_{t-1}[m_t] + (2 - \phi)\frac{1}{2}\left(p_t^1 + p_t^2\right) + \mathcal{M}$$

$$p_t^2 = \mathbb{E}_{t-2}[p_t^*] = (\phi - 1)\mathbb{E}_{t-2}[m_t] + (2 - \phi)\frac{1}{2}\left(\mathbb{E}_{t-2}[p_t^1] + p_t^2\right) + \mathcal{M}$$

- p_t^2 is observed in period $t-1$, so no expectation needed
- Price setters don't need expectations about their own prices

Rearrange and solve

$$p_t^1 = \frac{2(\phi-1)}{\phi}\mathbb{E}_{t-1}[m_t] + \frac{(2-\phi)}{\phi}p_t^2 + \frac{2}{\phi}\mathcal{M}$$

$$p_t^2 = \frac{2(\phi-1)}{\phi}\mathbb{E}_{t-2}[m_t] + \frac{(2-\phi)}{\phi}\mathbb{E}_{t-2}[p_t^1] + \frac{2}{\phi}\mathcal{M}$$

- Rational expectations imply that resetters 2 periods ago knew the other's policy function $\implies$ plug in + some tedious algebra

$$p_t^2 = \mathbb{E}_{t-2}[m_t] + \frac{1}{(\phi-1)}\mathcal{M}$$

- Exactly the same as before: base prices on expected demand
- Use this expression to solve for p_t^2

$$p_t^1 = \mathbb{E}_{t-1}[m_t] + \frac{\phi-2}{\phi}\underbrace{(\mathbb{E}_{t-1}[m_t] - \mathbb{E}_{t-2}[m_t])}_{\text{Updated information set}} + \frac{1}{(\phi-1)}\mathcal{M}$$

Equilibrium with Fischer contracts

$$
\begin{aligned}
p_t &= \frac{1}{2}\left(p_t^1 + p_t^2\right) \\
&= \frac{1}{2}\left(\left[\mathbb{E}_{t-1}[m_t] + \frac{\phi-2}{\phi}\left(\mathbb{E}_{t-1}[m_t] - \mathbb{E}_{t-2}[m_t]\right)\right] + \left[\mathbb{E}_{t-2}[m_t]\right]\right) + \frac{1}{(\phi-1)}\mathcal{M} \\
&= \mathbb{E}_{t-1}[m_t] - \frac{1}{\phi}\left(\mathbb{E}_{t-1}[m_t] - \mathbb{E}_{t-2}[m_t]\right) + \frac{1}{(\phi-1)}\mathcal{M} \\
y_t &= m_t - \mathbb{E}_{t-1}[m_t] + \frac{1}{\phi}\left(\mathbb{E}_{t-1}[m_t] - \mathbb{E}_{t-2}[m_t]\right) - \frac{1}{(\phi-1)}\mathcal{M} \\
&= \varepsilon_t + \frac{1}{\phi}\varepsilon_{t-1} - \frac{1}{(\phi-1)}\mathcal{M}
\end{aligned}
$$

Consider a surprising increase in m_t, announced after p_t^2 was set

Equilibrium with Fischer contracts

$$
\begin{aligned}
p_t &= \frac{1}{2}\left(p_t^1 + p_t^2\right) \\
&= \frac{1}{2}\left(\left[\mathbb{E}_{t-1}[m_t] + \frac{\phi-2}{\phi}\left(\mathbb{E}_{t-1}[m_t] - \mathbb{E}_{t-2}[m_t]\right)\right] + \left[\mathbb{E}_{t-2}[m_t]\right]\right) + \frac{1}{(\phi-1)}\mathcal{M} \\
&= \mathbb{E}_{t-1}[m_t] - \frac{1}{\phi}\left(\mathbb{E}_{t-1}[m_t] - \mathbb{E}_{t-2}[m_t]\right) + \frac{1}{(\phi-1)}\mathcal{M} \\
y_t &= m_t - \mathbb{E}_{t-1}[m_t] + \frac{1}{\phi}\left(\mathbb{E}_{t-1}[m_t] - \mathbb{E}_{t-2}[m_t]\right) - \frac{1}{(\phi-1)}\mathcal{M} \\
&= \varepsilon_t + \frac{1}{\phi}\varepsilon_{t-1} - \frac{1}{(\phi-1)}\mathcal{M}
\end{aligned}
$$

Consider a surprising increase in m_t, announced after p_t^2 was set

- Prices are **lower** compared to flexprice: p^2s were locked in in $t-1$ at too low level
- Output is **higher** compared to flexprice
- If labor is inelastic the costs of fixing prices are lower

Monetary policy

Thinking of m as completely random seems strange

- Central banks are not trying to surprise anybody
- They try to "lean against the wind" and work against demand shocks

New definition of aggregate demand

- Postulate the following relationship:

$$y_t = m_t^{cb} - p_t + v_t$$

- New definition: v_t represents shocks to aggregate demand
- Think of m_t^{cb} as (potentially active) monetary policy
- Important: monetary policy does not know more than the market

New equilibrium

Note: m_t^{cb} and v_t always enter as a sum $\implies$ substitute for m_t

$$p_t = \mathbb{E}_{t-1}[m_t^{cb} + v_t] - \frac{1}{\phi}\left(\mathbb{E}_{t-1}[m_t^{cb} + v_t] - \mathbb{E}_{t-2}[m_t^{cb} + v_t]\right) + \frac{1}{(\phi - 1)}\mathcal{M}$$

$$y_t = m_t^{cb} + v_t - \mathbb{E}_{t-1}[m_t^{cb} + v_t] + \frac{1}{\phi}\left(\mathbb{E}_{t-1}[m_t^{cb} + v_t] - \mathbb{E}_{t-2}[m_t^{cb} + v_t]\right) - \frac{1}{(\phi - 1)}\mathcal{M}$$

Question

- If v_t is random, but m_t^{cb} can be controlled, what should policy makers do?
- What's the **optimal** monetary policy to stabilize output?

Fischer contracts with demand stabilization

Environment

- Assume v_t follows a random walk: $v_t = v_{t-1} + \varepsilon_t$ with $\varepsilon_t \sim N(0, \sigma_\varepsilon^2)$
- Assume monetary policy is given by

$$m_t^{cb} = a_1 \varepsilon_{t-1} + a_2 \varepsilon_{t-2} + \cdots$$

- Policy makers know all past information
- Policy rule is perfectly credible but cannot react immediately
- The policy rule only contains linear terms – this turns out to be enough, but we will return to this issue.
- Question: What should be the weight on each a_x term?

Fischer contracts with demand stabilization II

Rewriting

- Plugging in the information on the previous slide gives

$$m_t^{cb} + v_t = \underbrace{a_1\varepsilon_{t-1} + a_2\varepsilon_{t-2} + \cdots}_{m_t} + \underbrace{v_{t-1} + \varepsilon_t}_{v_t}$$

- This is a model with rational expectations, hence the policy rule enters expectations

$$\begin{aligned} y_t =& m_t^{cb} + v_t - \mathbb{E}_{t-1}\left[m_t^{cb} + v_t\right] \\ &+ \frac{1}{\phi}\left(\mathbb{E}_{t-1}\left[m_t^{cb} + v_t\right] - \mathbb{E}_{t-2}\left[m_t^{cb} + v_t\right]\right) - \frac{1}{(\phi - 1)}\mathcal{M} \end{aligned}$$

Fischer contracts with demand stabilization III

$$
\begin{aligned}
y_t =& m_t^{cb} + v_t - \mathbb{E}_{t-1}\left[m_t^{cb} + v_t\right] \\
&+ \frac{1}{\phi}\left(\mathbb{E}_{t-1}\left[m_t^{cb} + v_t\right] - \mathbb{E}_{t-2}\left[m_t^{cb} + v_t\right]\right) - \frac{1}{(\phi - 1)}\mathcal{M}
\end{aligned}
$$

Future can be expressed in terms of the past

$$
\begin{aligned}
\mathbb{E}_{t-1}\left[m_t^{cb} + v_t\right] &= \mathbb{E}_{t-1}\left[a_1\varepsilon_{t-1} + a_2\varepsilon_{t-2} + a_3\varepsilon_{t-3} + \cdots + v_{t-1} + \varepsilon_t\right] \\
&= m_t^{cb} + v_{t-1} + \mathbb{E}_{t-1}\left[\varepsilon_t\right] \\
&= m_t^{cb} + v_{t-1} \\
\mathbb{E}_{t-2}\left[m_t^{cb} + v_t\right] &= \mathbb{E}_{t-2}\left[a_1\varepsilon_{t-1} + a_2\varepsilon_{t-2} + a_3\varepsilon_{t-3} + \cdots + v_{t-2} + \varepsilon_{t-1} + \varepsilon_t\right] \\
&= \mathbb{E}_{t-2}\left[a_1\varepsilon_{t-1} + a_2\varepsilon_{t-2} + a_3\varepsilon_{t-3} + \cdots + \varepsilon_{t-1} + \varepsilon_t\right] + v_{t-2} \\
&= \mathbb{E}_{t-2}\left[a_1\varepsilon_{t-1} + \varepsilon_{t-1} + \varepsilon_t\right] + a_2\varepsilon_{t-2} + a_3\varepsilon_{t-3} + \cdots + v_{t-2} \\
&= a_2\varepsilon_{t-2} + a_3\varepsilon_{t-3} + \cdots + v_{t-2}
\end{aligned}
$$

Fischer contracts with demand stabilization IV

$$y_t = m_t^{cb} + v_t - \mathbb{E}_{t-1}[m_t^{cb} + v_t] + \frac{1}{\phi}\left(\mathbb{E}_{t-1}[m_t^{cb} + v_t] - \mathbb{E}_{t-2}[m_t^{cb} + v_t]\right) - \frac{1}{(\phi - 1)}\mathcal{M}$$

$$\mathbb{E}_{t-1}[m_t^{cb} + v_t] = m_t^{cb} + v_{t-1}$$

$$\mathbb{E}_{t-2}[m_t^{cb} + v_t] = a_2\varepsilon_{t-2} + a_3\varepsilon_{t-3} + \cdots + v_{t-2}$$

Deriving output with monetary policy

$$\begin{aligned}\mathbb{E}_{t-1}[m_t^{cb} + v_t] - \mathbb{E}_{t-2}[m_t^{cb} + v_t] &= m_t^{cb} - a_2\varepsilon_{t-2} - a_3\varepsilon_{t-3} - \cdots + v_{t-1} - v_{t-2} \\ &= a_1\varepsilon_{t-1} + \varepsilon_{t-1} = (1 + a_1)\varepsilon_{t-1}\end{aligned}$$

$$\begin{aligned}m_t^{cb} + v_t - \mathbb{E}_{t-1}[m_t^{cb} + v_t] &= m_t^{cb} + v_t - m_t^{cb} - v_{t-1} \\ &= v_t - v_{t-1} = \varepsilon_t\end{aligned}$$

$$y_t = \varepsilon_t + \frac{1 + a_1}{\phi}\varepsilon_{t-1} - \frac{1}{(\phi - 1)}\mathcal{M}$$

Optimal monetary policy

The central bank can affect output with a_1

$$y_t = \varepsilon_t + \frac{1 + a_1}{\phi}\varepsilon_{t-1} - \frac{1}{(\phi - 1)}\mathcal{M}$$

Other terms are irrelevant

- Contracts are only set for two periods
- The central bank (CB) cannot see what will happen tomorrow

$\Longrightarrow$ CB can eliminate effects of unanticipated shocks

Minimize variance of output

$$\begin{aligned}\mathrm{Var}(y_t) &= \mathrm{Var}(\varepsilon_t) + \frac{(1 + a_1)^2}{\phi^2}\mathrm{Var}(\varepsilon_{t-1}) \\ &= \sigma_\varepsilon^2 + \frac{(1 + a_1)^2}{\phi^2}\sigma_\varepsilon^2\end{aligned}$$

Optimal monetary policy

The central bank can affect output with a_1

$$y_t = \varepsilon_t + \frac{1 + a_1}{\phi}\varepsilon_{t-1} - \frac{1}{(\phi - 1)}\mathcal{M}$$

Other terms are irrelevant

- Contracts are only set for two periods
- The central bank (CB) cannot see what will happen tomorrow

$\Longrightarrow$ CB can eliminate effects of unanticipated shocks

Minimize variance of output

$$\begin{aligned}
\mathrm{Var}(y_t) &= \mathrm{Var}(\varepsilon_t) + \frac{(1 + a_1)^2}{\phi^2}\mathrm{Var}(\varepsilon_{t-1}) \\
&= \sigma_\varepsilon^2 + \frac{(1 + a_1)^2}{\phi^2}\sigma_\varepsilon^2 \\
&= \sigma_\varepsilon^2 \text{ with } a_1 = -1
\end{aligned}$$

Central bankers save firms from frictions

The central bank can return the economy equilibrium with less frictions

- It neutralizes all unanticipated **demand** shocks it observes
- Firms anticipate the actions of the CB
- Demand shocks still have effects, but only for one period

$\Longrightarrow$ as if contract length was only 1 period

Optimal policy

- If the policy goal is to minimize output volatility, a linear rule is enough
- "Lean against the wind": If aggregate demand is high, decrease money supply

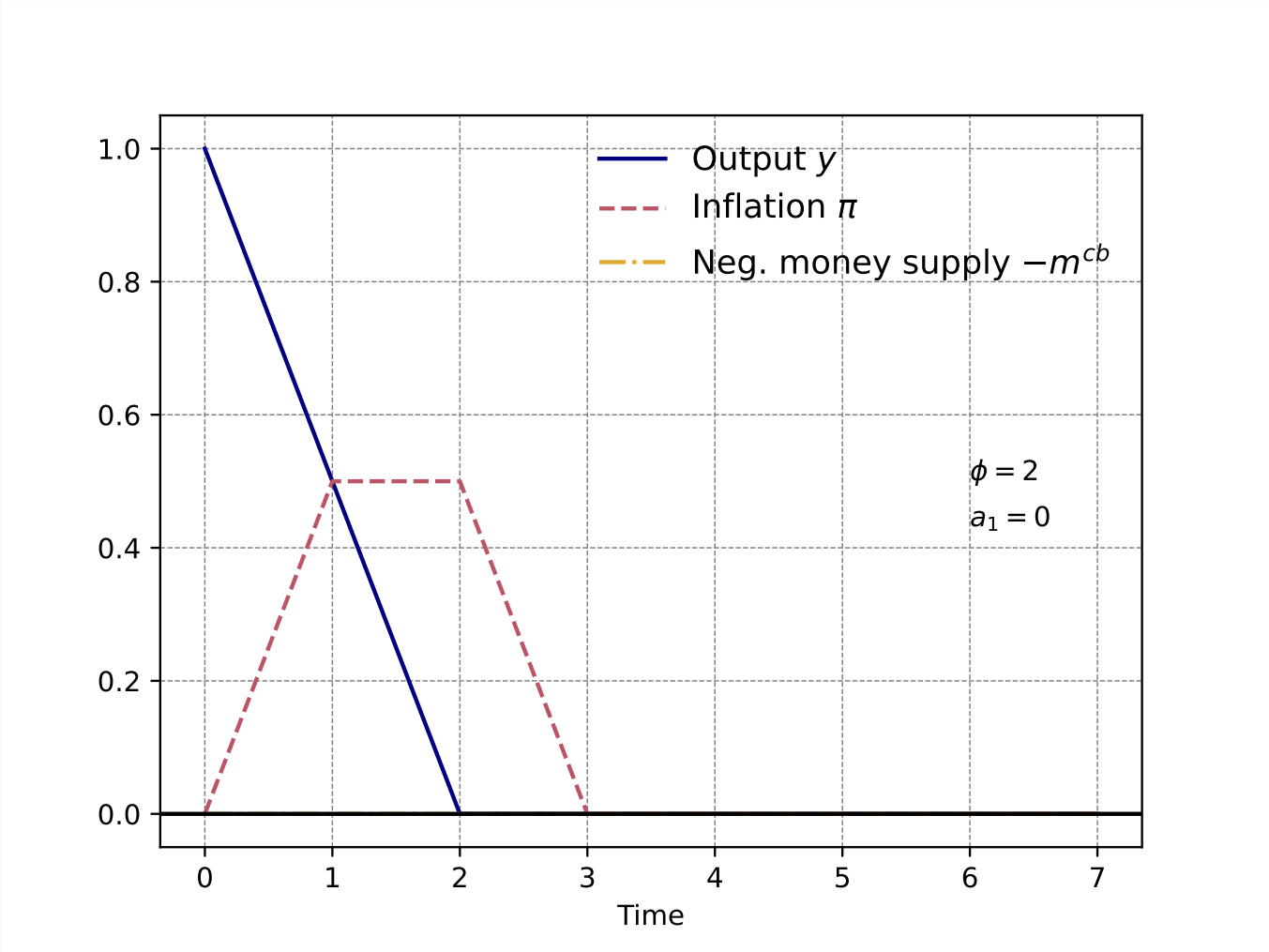
Summary – persistence of shocks

Fischer Contracts – m_t is random walk (restrict $a_{>1} = 0$)

$$
\begin{aligned}
p_t &= \mathbb{E}_{t-1}[m_t^{cb} + v_t] - \frac{1}{\phi}\left(\mathbb{E}_{t-1}[m_t^{cb} + v_t] - \mathbb{E}_{t-2}[m_t^{cb} + v_t]\right) \\
&= a_1\varepsilon_{t-1} + v_{t-2} + \varepsilon_{t-1} - \frac{1}{\phi}(1 + a_1)\varepsilon_{t-1} \\
&= v_{t-2} + \frac{\phi - 1}{\phi}(1 + a_1)\varepsilon_{t-1} \\
\implies p_t - p_{t-1} &= v_{t-2} + \frac{\phi - 1}{\phi}(1 + a_1)\varepsilon_{t-1} - \left(v_{t-3} + \frac{\phi - 1}{\phi}(1 + a_1)\varepsilon_{t-2}\right) \\
\pi_t &= \frac{\phi - 1}{\phi}(1 + a_1)\varepsilon_{t-1} - \frac{\phi - 1}{\phi}(1 + a_1)\varepsilon_{t-2} + \varepsilon_{t-2} \\
y_t &= \varepsilon_t + \frac{1 + a_1}{\phi}\varepsilon_{t-1} - \frac{1}{(\phi - 1)}\mathcal{M}
\end{aligned}
$$

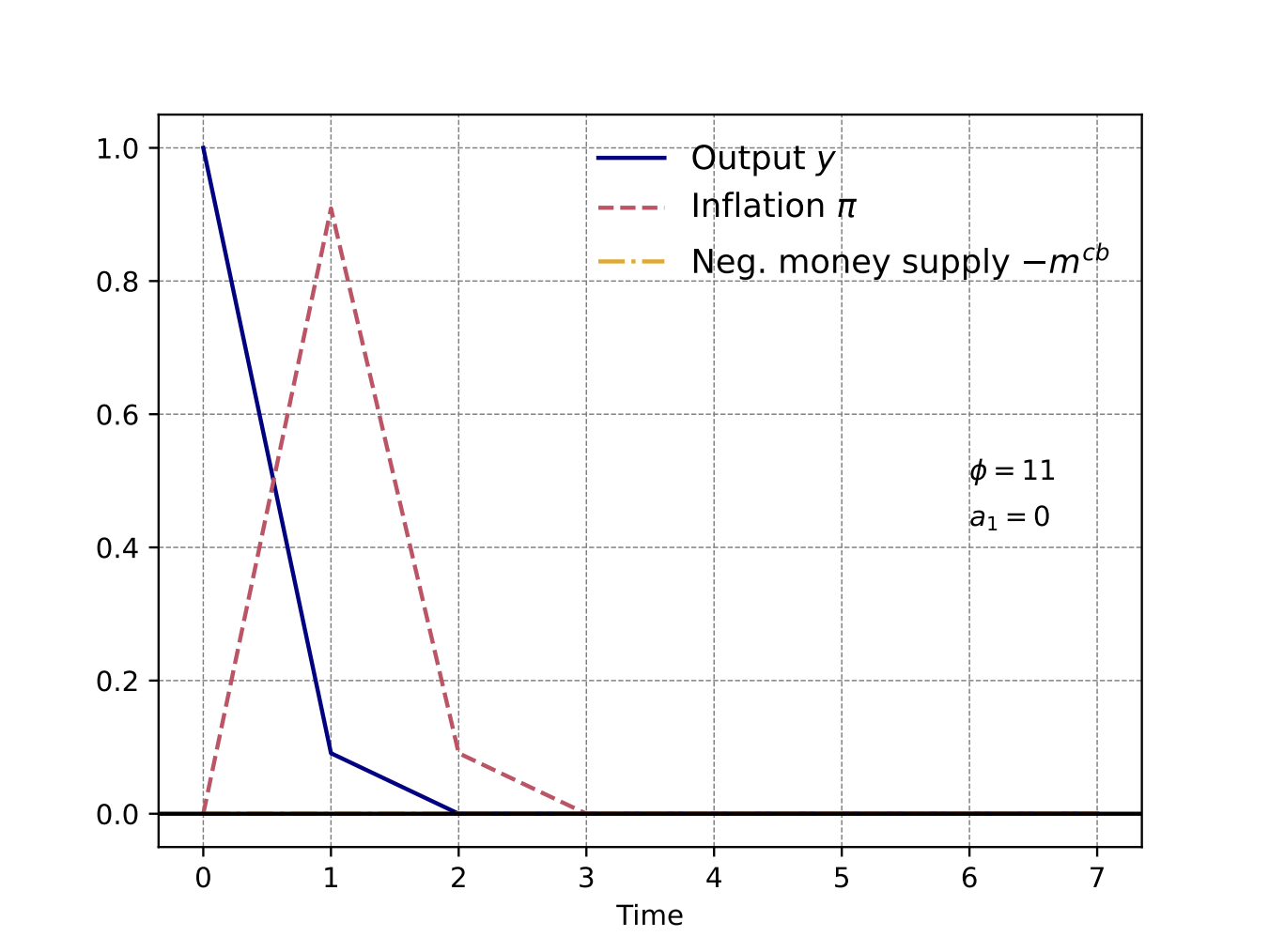
- Remember: prices are reset in $t-1$ for t and $t+1$ $\implies$ shock lasts two periods
- As labor becomes inelastic ($\phi \uparrow$), y_t moves less, π_t more

Optimal monetary policy in a picture



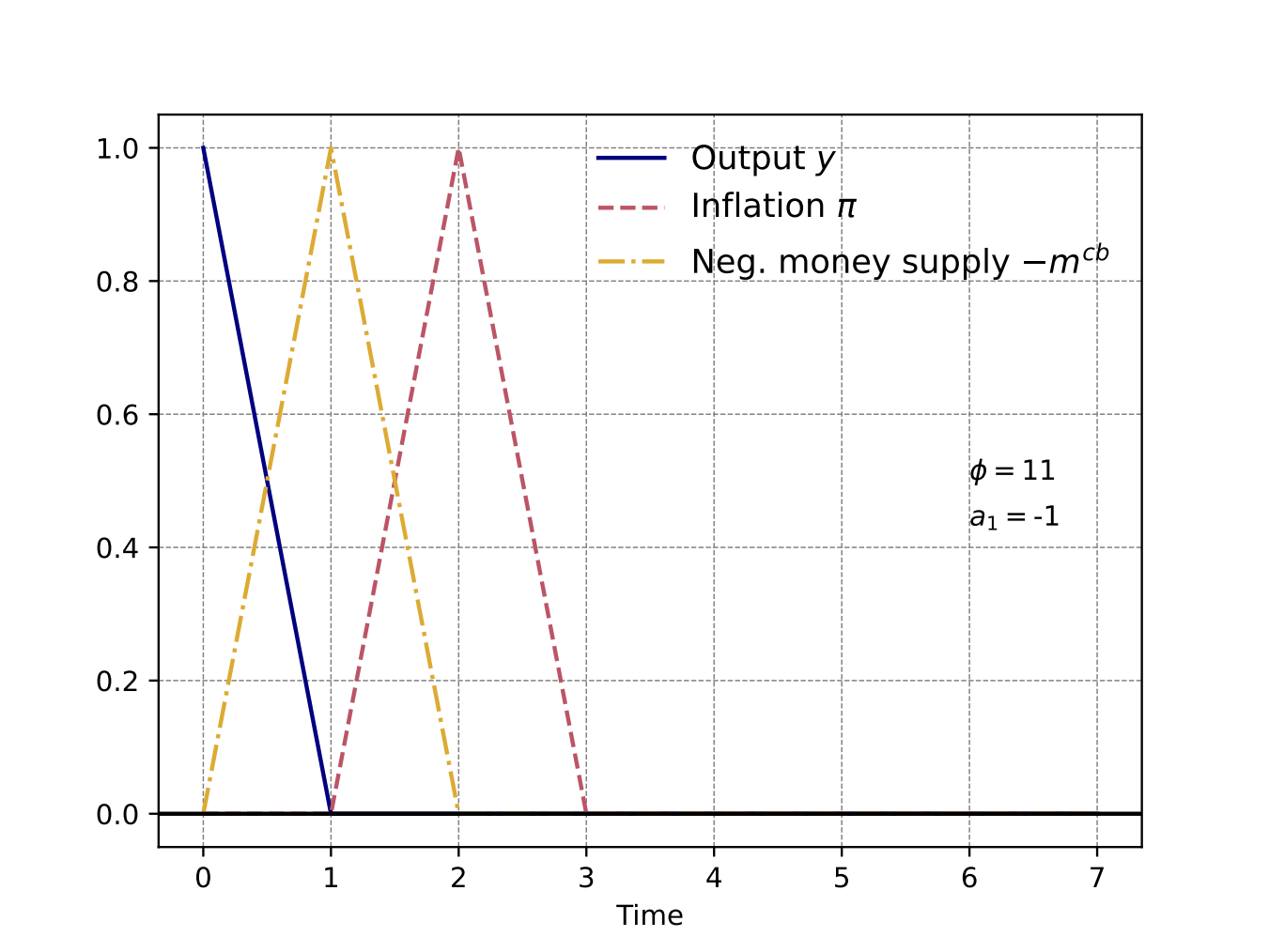
- $v_0 = 1 \implies$ demand $m = m^{cb} + v$ is **permanently higher** $(\rho = 1)$
- Since $y = m^{cb} + v - p$, we need $\sum \pi_t = 1$ for $y = 0$

Optimal monetary policy in a picture



- Main action in period 1, when first firms reset prices
- Prices absorb bigger share of shock with inelastic labor

Optimal monetary policy in a picture



- Monetary policy can suppress demand back to steady state
- Buys time for firms locked in, p changes later

Taylor pricing (Time-dependent)

- Not all firms can change prices each period. Only a fixed fraction gets a "reset opportunity" each period
- When a firm *does* reset, it chooses a price that will remain in place for a **fixed duration** (e.g. 12 months)
- Resetting firms observe current conditions but know many rivals are still stuck at old prices
  → they adjust only partway today, expecting further alignment as others reset later
- Because only a subset adjusts each period, the aggregate price level updates *gradually*

Menu Costs (State-dependent) and the (S,s) Band

- Changing prices is not free, but requires **fixed costs**
  (printing menus, relabeling, managerial costs, confusing customers)
- With a small menu cost, the firm tolerates small deviations between the actual and the desired (frictionless) price p_t^*:

$$|p_t - p_t^*| \leq \text{(S,s)-band} \quad \Rightarrow \quad \text{keep price unchanged}$$

  → (S,s)-band logic applies in many other instances
  (e.g. fixed capital adjustment costs)
- Only when shocks move the desired price far enough so that $|p_t - p_t^*|$ exceeds the band, the firm pays the menu cost and resets p_t

Result:

- Prices change infrequently but in larger "jumps"
- Price setting depends on the state of the firm / economy
  (state-dependent friction)

Price adjustment in menu cost and Calvo models

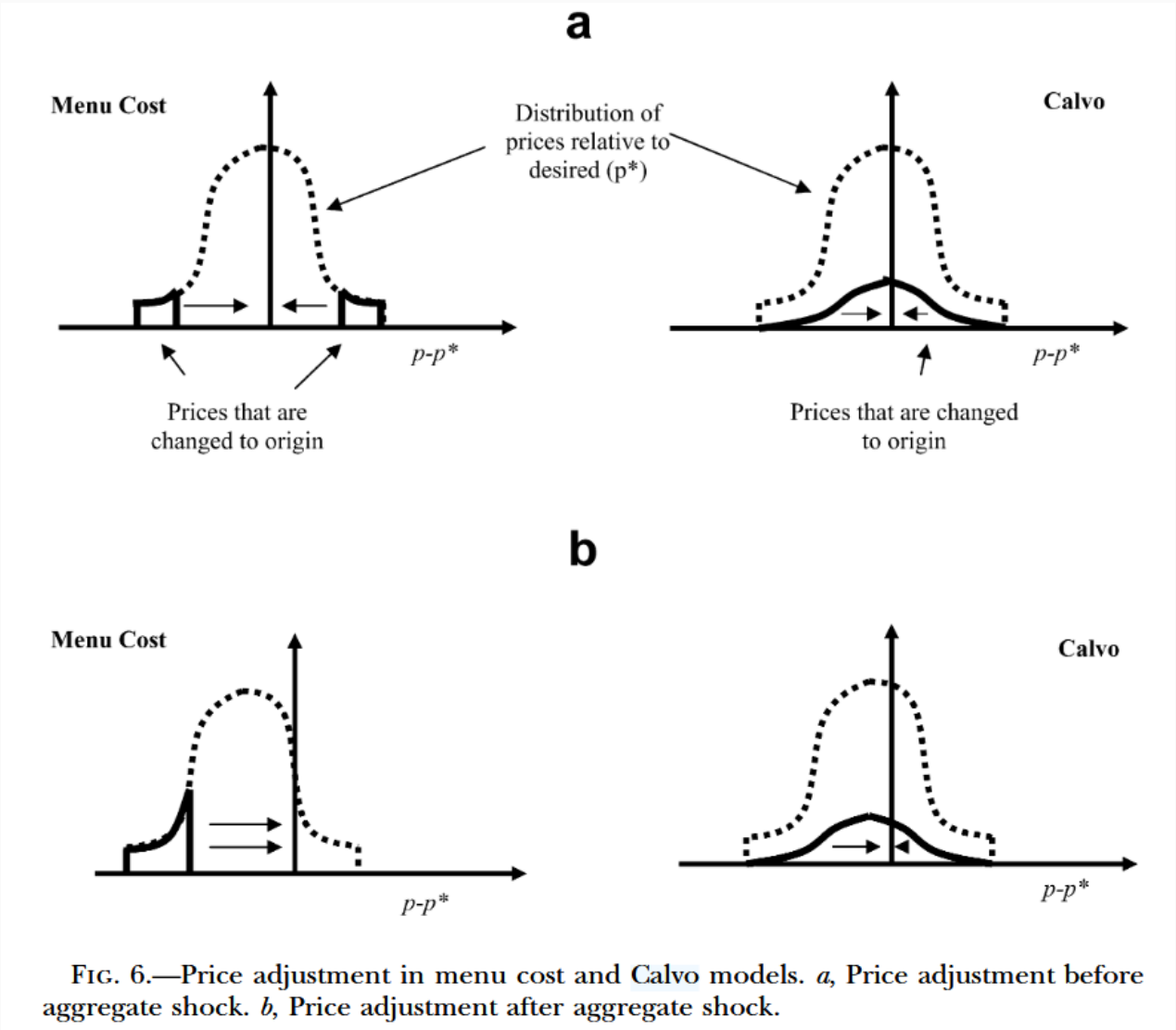


FIG. 6.—Price adjustment in menu cost and Calvo models. *a*, Price adjustment before aggregate shock. *b*, Price adjustment after aggregate shock.

Golosov-Lucas 2007

Punchline

Pricing frictions

- Romer's conclusion carries through: without frictions, anticipated shocks have no effect on output
- If agents cannot adjust prices freely, however, demand shocks have real effects
- Monetary policy can help stabilize output

Different forms of rigidity

- Fischer contracts: set price schedule and stick to it
- Taylor contracts: set prices for fixed number of periods
- Menu costs: state-dependent, adjust only when sufficiently far from the desired price

Next time

- Most influential form of price rigidity: the Calvo fairy